

Numpy Statistics

LAB(5)

Part-3

1. Student Exam Scores Analysis

- Objective: Analyze score distributions using dot plots, histograms, and z-scores.
- Data: Simulated exam scores (mean=70, std=10) for 100 students.

Tasks:

- Generate data with `np.random.normal()`.
- Create a dot plot (individual scores) and histogram (score distribution).
- Calculate z-scores for each student.
- Identify outliers (z-score > 2 or < -2).

Visualizations:

Dot plot: `plt.scatter()` with jitter.

Histogram: `plt.hist()`.

```
# Generate data
np.random.seed(42)
scores = np.random.normal(70, 10, 100)
```

2. Fruit Sales Distribution

- Objective: Visualize categorical data with a pie chart.
- Data: Sales quantities for 5 fruits (apples, bananas, oranges, grapes, strawberries).
- Tasks:

- Generate random sales data with `np.random.randint()`.
- Create a pie chart showing sales proportions.
- Highlight the highest-selling fruit.

Visualization:

Pie chart: `plt.pie()` with labels and explode.

```
fruits = ['Apples', 'Bananas', 'Oranges', 'Grapes', 'Strawberries']
sales = np.random.randint(50, 200, size=5)
explode = [0.1 if s == max(sales) else 0 for s in sales] # Highlight highest
```

3. Temperature vs. Ice Cream Sales

- Objective: Investigate correlation using scatter plots and Pearson's R.
- Data: Daily temperature (°C) and ice cream sales (\$) for 30 days.
- Tasks:
 - Simulate correlated data with `np.corrcoef()`.
 - Create a scatter plot with regression line.
 - Calculate Pearson's R and interpret correlation strength.

Visualizations:

Scatter plot: `plt.scatter()`.

Regression line: `np.polyfit()` + `plt.plot()`.

```
# Generate correlated data
np.random.seed(42)
temp = np.linspace(20, 35, 30)
sales = 50 + 2 * temp + np.random.normal(0, 5, 30)
```

4. Income Distribution Across Professions

- Objective: Compare distributions with box plots.
- Data: Simulated incomes for 3 professions (engineers, teachers, artists).
- Tasks:
 - Generate skewed income data (`np.random.exponential()`).
 - Create side-by-side box plots for each profession.
 - Identify median, IQR, and outliers.

Visualization:

Box plot: `plt.boxplot()`.

```
# Generate income data
np.random.seed(42)
engineers = np.random.exponential(80000, 100) + 40000
teachers = np.random.normal(55000, 8000, 100)
artists = np.random.lognormal(10.5, 0.4, 100)
```

5. Study Hours vs. Exam Scores Regression

- Objective: Perform linear regression analysis.
- Data: Study hours and exam scores for 50 students.
- Tasks:
 - Simulate linear data: $\text{scores} = 2 * \text{hours} + \text{noise}$.
 - Fit a regression line using `np.polyfit()`.
 - Plot scatter plot + regression line.

- Calculate and interpret R^2 .

```
# Generate data
np.random.seed(42)
hours = np.random.uniform(1, 10, 50)
scores = 30 + 7 * hours + np.random.normal(0, 5, 50)
```

6. Website Traffic Analysis

- Objective: Analyze traffic distribution with histograms and z-scores.
- Data: Hourly website visits for 1 month (Poisson-distributed).
- Tasks:
 - Generate data: `np.random.poisson(lam=50)`.
 - Plot histogram of visits.
 - Calculate z-scores for peak traffic hours.
 - Identify anomalies (z-score > 3).

```
# Generate data
np.random.seed(42)
traffic = np.random.poisson(lam=50, size=744) # 31 days * 24 hours
```

7. Advertising Spend vs. Revenue

- Objective: Quantify correlation and visualize with a scatter plot.
- Data: Monthly ad spend (\$) and revenue (\$) for 12 months.
- Tasks:
 - Simulate data with strong positive correlation.
 - Create a scatter plot.
 - Compute Pearson's R and p-value (`scipy.stats.pearsonr`).
 - Annotate the plot with R value.

```
# Generate correlated data
np.random.seed(42)
spend = np.linspace(1000, 10000, 12)
revenue = 5000 + 2.5 * spend + np.random.normal(0, 1000, 12)
```

8. Movie Genre Popularity

- Objective: Visualize categorical proportions with a pie chart.
- Data: Ticket sales for 6 movie genres.
- Tasks:
 - Generate random sales data.

- Create a pie chart with percentage labels.
- Highlight the least popular genre.

```
genres = ['Action', 'Comedy', 'Drama', 'Horror', 'Sci-Fi', 'Romance']
tickets = np.random.randint(5000, 30000, size=6)
min_idx = np.argmin(tickets)
explode = [0.1 if i == min_idx else 0 for i in range(len(genres))]
```

9. Housing Price Analysis

- Objective: Compare price distributions across neighborhoods with box plots.
- Data: Simulated house prices for 3 neighborhoods (normal/log-normal).
- Tasks:
 - Generate data with different means/variances.
 - Create vertical box plots for each neighborhood.
 - Calculate and compare medians/IQRs.

```
# Generate neighborhood data
np.random.seed(42)
suburban = np.random.normal(350000, 50000, 100)
urban = np.random.lognormal(12.8, 0.3, 100)
rural = np.random.exponential(250000, 100) + 150000
```

10. Athlete Performance: Age vs. Speed

- Objective: Analyze correlation and regression.
- Data: Age (years) and sprint speed (m/s) for 40 athletes.
- Tasks:
 - Simulate negative correlation: $\text{speed} = 10 - 0.1 \cdot \text{age} + \text{noise}$.
 - Plot scatter plot + regression line.
 - Calculate Pearson's R and discuss the relationship.

```
# Generate data with negative correlation
np.random.seed(42)
age = np.random.randint(18, 40, 40)
speed = 10 - 0.15 * age + np.random.normal(0, 0.5, 40)
```